Correction to: Electrochemical Impedance Spectroscopy and its Applications

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Correction to: A. Lasia, *Electrochemical Impedance Spectroscopy and its Applications*

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Unfortunately the original version of the book had been published with several incorrect equations. The corrected equations have been listed below.

p. 39, Eq. (2.113)

$$\begin{aligned} i(t) &= \frac{E_0}{R}\frac{1}{1+\dfrac{1}{(\omega RC)^2}}\left[\tan(\varphi)\cos(\omega t)+\sin(\omega t)\right] = \\ &= \frac{E_0}{R}\cos(\varphi)\left[\frac{\cos(\omega t)\sin(\varphi)}{\cos(\varphi)}+\sin(\omega t)\right] \\ &= \frac{E_0}{R}\left[\cos(\omega t)\sin(\varphi)+\sin(\omega t)\cos(\varphi)\right] \end{aligned} \tag{2.113}$$

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A. Lasia, *Electrochemical Impedance Spectroscopy and its Applications*,
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p. 40. Eq. (2.118)

$$\begin{aligned} i(t) &= \frac{E_0}{R}\frac{1}{1+\frac{1}{(\omega RC)^2}}\left[\cos(\omega t) - \tan(\varphi)\sin(\omega t)\right] = \\ &= \frac{E_0}{R}\frac{1}{1+\frac{1}{(\omega RC)^2}}\left[\frac{\cos(\omega t)\cos(\omega t) - \sin(\varphi)\sin(\omega t)}{\cos(\varphi)}\right] \end{aligned} \tag{2.118}$$

p. 46, Eq. (2.135)–(2.136)

$$\begin{aligned} \widehat{Z}(j\omega) &= R_0 + \frac{R_1}{1+j\omega R_1 C_1} = R_0 + \frac{R_1(1-j\omega R_1 C_1)}{1+(\omega R_1 C_1)^2} = \\ &= R_0 + \frac{R_1}{1+(\omega R_1 C_1)^2} - j\frac{\omega R_1^2 C_1}{1+(\omega R_1 C_1)^2} \end{aligned} \tag{2.135}$$

$$\begin{aligned} \mathrm{Re}\left[\widehat{Z}(j\omega)\right] &= R_0 + \frac{R_1}{1+(\omega R_1 C_1)^2} \\ \mathrm{Im}\left[\widehat{Z}(j\omega)\right] &= -\frac{\omega R_1^2 C_1}{1+(\omega R_1 C_1)^2} \end{aligned} \tag{2.136}$$

p. 50, Eq. (2.140)

$$\begin{aligned} |Z| &= \sqrt{R^2 + \frac{1}{(\omega C)^2}} \\ \varphi &= \mathrm{atan}\left(\frac{Z''}{Z'}\right) = \mathrm{atan}\left(-\frac{1}{\omega RC}\right) = -\mathrm{atan}\left(\frac{1}{\omega RC}\right) \end{aligned} \tag{2.140}$$

p. 92, Eq. (4.39).

$$C_{\mathrm{O}}(0) = 1 - \frac{i}{i_{\mathrm{lim}}} \text{ and } C_{\mathrm{R}}(0) = \frac{i}{i_{\mathrm{lim}}} \tag{4.39}$$

p. 92, Eq. (4.41)

$$\begin{aligned} \widehat{Z}_{\mathrm{W}} &= \widehat{Z}_{\mathrm{W,O}} + \widehat{Z}_{\mathrm{W,R}} = \frac{RT}{n^2F^2\sqrt{j\omega}}\left(\frac{1}{\sqrt{D_{\mathrm{O}}}C_{\mathrm{O}}(0)} + \frac{1}{\sqrt{D_{\mathrm{R}}}C_{\mathrm{R}}(0)}\right) \\ &= \frac{\sigma'}{\sqrt{j\omega}} = \frac{\sqrt{2}}{\sqrt{j}}\frac{\sigma}{\sqrt{\omega}} = \sigma(1-j) \end{aligned} \tag{4.41}$$

p. 105, Eq. (4.76)

$$\lim_{\omega\to 0}\left(\widehat{Z}_{\mathrm{t}}\right) = R_{\mathrm{S}} + R_{\mathrm{ct}} + \frac{\sigma' l}{\sqrt{D_{\mathrm{O}}}} \tag{4.76}$$

p. 113, Eq. (4.106)

$$\begin{aligned}\widehat{Z}_{\mathrm{W}} &= \frac{RT}{n^2F^2\sqrt{D_{\mathrm{R}}}C_{\mathrm{R}}(0)}\frac{1}{\sqrt{j\omega}\coth\left(\sqrt{\frac{j\omega}{D_{\mathrm{R}}}}r_0\right) - \frac{\sqrt{D_{\mathrm{R}}}}{r_0}} = \\ &= \frac{\sigma'}{\sqrt{j\omega}}\frac{1}{\coth\left(\sqrt{\frac{j\omega}{D_{\mathrm{R}}}}r_0\right) - \frac{\sqrt{D_{\mathrm{R}}}}{r_0\sqrt{j\omega}}} = \\ &= \frac{\sigma' r_0}{\sqrt{D_{\mathrm{R}}}}\frac{1}{\left(\sqrt{\frac{j\omega}{D_{\mathrm{R}}}}r_0\right)\coth\left(\sqrt{\frac{j\omega}{D_{\mathrm{R}}}}r_0\right) - 1}\end{aligned} \tag{4.106}$$

p. 141, Eq. (5.71)

$$\mathrm{B_{ads}} + \mathrm{e} \underset{\overleftarrow{k}_{-2}}{\overset{\overrightarrow{k}_{2}}{\rightleftharpoons}} \mathrm{C_{ads}} \tag{5.71}$$

p. 142, Eq. (5.80)

$$\widetilde{i} = -F\left[\left(\frac{\partial r_0}{\partial \eta}\right)_{\theta_{\mathrm{B}},\theta_{\mathrm{C}}}\widetilde{\eta} + \left(\frac{\partial r_0}{\partial \theta_{\mathrm{B}}}\right)_{\eta,\theta_{\mathrm{C}}}\widetilde{\theta}_{\mathrm{B}} + \left(\frac{\partial r_0}{\partial \theta_{\mathrm{C}}}\right)_{\eta,\theta_{\mathrm{B}}}\widetilde{\theta}_{\mathrm{C}}\right] \tag{5.80}$$

p. 150, Eq. (6.19), (6.20), (6.21) (errors in signs)

$$\begin{aligned}-\left(\frac{\partial r_0}{\partial \eta}\right) &= \frac{1}{F}\frac{\widetilde{i}}{\widetilde{\eta}} + \left(\frac{\partial r_0}{\partial \theta_{\mathrm{B}}}\right)\frac{\widetilde{\theta}_{\mathrm{B}}}{\widetilde{\eta}} + \left(\frac{\partial r_0}{\partial C_{\mathrm{A}}}\right)\frac{\widetilde{C}_{\mathrm{A}}(0)}{\widetilde{\eta}} + \left(\frac{\partial r_0}{\partial C_{\mathrm{C}}}\right)\frac{\widetilde{C}_{\mathrm{C}}(0)}{\widetilde{\eta}} \\ 0 &= -\frac{1}{2F}\frac{\widetilde{i}}{\widetilde{\eta}} + \sqrt{j\omega D_{\mathrm{A}}}\frac{\widetilde{C}_{\mathrm{A}}(0)}{\widetilde{\eta}} \\ 0 &= \frac{1}{2F}\frac{\widetilde{i}}{\widetilde{\eta}} + \sqrt{j\omega D_{\mathrm{C}}}\frac{\widetilde{C}_{\mathrm{C}}(0)}{\widetilde{\eta}} \\ -\left(\frac{\partial r_1}{\partial \eta}\right) &= -\Gamma_{\infty} j\omega\frac{\widetilde{\theta}_{\mathrm{B}}}{\widetilde{\eta}} + \left(\frac{\partial r_1}{\partial \theta_{\mathrm{B}}}\right)\frac{\widetilde{\theta}_{\mathrm{B}}}{\widetilde{\eta}} + \left(\frac{\partial r_1}{\partial C_{\mathrm{A}}}\right)\frac{\widetilde{C}_{\mathrm{A}}(0)}{\widetilde{\eta}} + \left(\frac{\partial r_1}{\partial C_{\mathrm{C}}}\right)\frac{\widetilde{C}_{\mathrm{C}}(0)}{\widetilde{\eta}}\end{aligned} \tag{6.19}$$

$$\begin{bmatrix} -\dfrac{\partial r_0}{\partial \eta} \\ 0 \\ 0 \\ -\dfrac{\partial r_1}{\partial \eta} \end{bmatrix} = \begin{bmatrix} \dfrac{1}{F} & \dfrac{\partial r_0}{\partial \Theta} & \dfrac{\partial r_0}{\partial C_A} & \dfrac{\partial r_0}{\partial C_C} \\ \dfrac{1}{2F} & 0 & \sqrt{j\omega D_A} & 0 \\ \dfrac{1}{2F} & 0 & 0 & \sqrt{j\omega D_C} \\ 0 & \dfrac{\partial r_1}{\partial \Theta} - \Gamma_\infty j\omega & \dfrac{\partial r_1}{\partial C_A} & \dfrac{\partial r_1}{\partial C_C} \end{bmatrix} \begin{bmatrix} \dfrac{\widetilde{i}}{\widetilde{\eta}} \\ \dfrac{\widetilde{\Theta}}{\widetilde{\eta}} \\ \dfrac{\widetilde{C}_C(0)}{\widetilde{\eta}} \\ \dfrac{\widetilde{C}_A(0)}{\widetilde{\eta}} \end{bmatrix} \tag{6.20}$$

$$A = \begin{vmatrix} \dfrac{1}{F} & \dfrac{\partial r_0}{\partial \theta_\mathrm{B}} & \dfrac{\partial r_0}{\partial C_\mathrm{A}} & \dfrac{\partial r_0}{\partial C_\mathrm{C}} \\ \dfrac{1}{-2F} & 0 & \sqrt{j\omega D_\mathrm{A}} & 0 \\ \dfrac{1}{2F} & 0 & 0 & \sqrt{j\omega D_C} \\ 0 & \dfrac{\partial r_1}{\partial \theta_\mathrm{B}} - \Gamma_\infty j\omega & \dfrac{\partial r_1}{\partial C_\mathrm{A}} & \dfrac{\partial r_1}{\partial C_\mathrm{C}} \end{vmatrix} \tag{6.21}$$

p. 153, Eq. (6.22), (6.23), (6.24)

$$T = \begin{vmatrix} -\dfrac{\partial r_0}{\partial \eta} & \dfrac{\partial r_0}{\partial \theta_\mathrm{B}} & \dfrac{\partial r_0}{\partial C_\mathrm{A}} & \dfrac{\partial r_0}{\partial C_\mathrm{C}} \\ 0 & 0 & \sqrt{j\omega D_\mathrm{A}} & 0 \\ 0 & 0 & 0 & \sqrt{j\omega D_\mathrm{C}} \\ -\dfrac{\partial r_1}{\partial \eta} & \dfrac{\partial r_1}{\partial \theta_\mathrm{B}} - \Gamma_\infty j\omega & \dfrac{\partial r_1}{\partial C_\mathrm{A}} & \dfrac{\partial r_1}{\partial C_\mathrm{C}} \end{vmatrix} \tag{6.22}$$

$$\begin{aligned} T &= \sqrt{D_\mathrm{A} D_\mathrm{C}} \left[\Gamma_\infty \frac{\partial r_0}{\partial \eta} (j\omega)^2 + \left(-\frac{\partial r_0}{\partial \eta}\frac{\partial r_1}{\partial \theta} + \frac{\partial r_0}{\partial \theta}\frac{\partial r_1}{\partial \eta} \right)(j\omega) \right] \\ &= a_4 (j\omega)^2 + a_2 (j\omega) \end{aligned} \tag{6.23}$$

$$\begin{aligned} A &= -\frac{1}{2F} \begin{bmatrix} 2\sqrt{D_\mathrm{A} D_\mathrm{C}} \Gamma_\infty (j\omega)^2 \\ +\Gamma_\infty \left(\sqrt{D_\mathrm{C}} \dfrac{\partial r_0}{\partial C_\mathrm{A}} - \sqrt{D_\mathrm{A}} \dfrac{\partial r_0}{\partial C_\mathrm{C}} \right)(j\omega)^{3/2} \\ -2\sqrt{D_A D_C} \dfrac{\partial r_1}{\partial \theta}(j\omega) \\ + \begin{pmatrix} -\sqrt{D_C} \dfrac{\partial r_0}{\partial C_\mathrm{A}} \dfrac{\partial r_1}{\partial \theta} + \sqrt{D_A} \dfrac{\partial r_0}{\partial C_\mathrm{C}} \dfrac{\partial r_1}{\partial \theta} \\ +\sqrt{D_C} \dfrac{\partial r_0}{\partial \theta} \dfrac{\partial r_1}{\partial C_\mathrm{A}} - \sqrt{D_A} \dfrac{\partial r_0}{\partial \theta} \dfrac{\partial r_1}{\partial C_\mathrm{C}} \end{pmatrix} (j\omega)^{1/2} \end{bmatrix} \\ &= \frac{1}{2F} \left[b_4 (j\omega)^2 + b_3 (j\omega)^{3/2} + b_2 (j\omega) + b_1 (j\omega)^{1/2} \right] \end{aligned} \tag{6.24}$$

p. 152, Eq. (6.27)

$$\begin{aligned}\widehat{Z}_{\mathrm{f}} &= \frac{1}{2F}\frac{b_4}{a_4} + \frac{1}{2F}\frac{b_4}{a_4}\left[\frac{\frac{b_3}{b_4}(j\omega) + \left(\frac{b_2}{b_4} - \frac{a_2}{a_4}\right)(j\omega)^{1/2} + \frac{b_1}{b_4}}{(j\omega)^{3/2} + \frac{a_2}{a_4}(j\omega)^{1/2}}\right] \\ &= R_{\mathrm{ct}} + \frac{1}{2F}\left[\frac{\frac{b_3}{a_4}(j\omega) + \left(\frac{b_2}{a_4} - \frac{a_2 b_4}{a_4^2}\right)(j\omega)^{1/2} + \frac{b_1}{a_4}}{(j\omega)^{1/2}\left[(j\omega) + \frac{a_2}{a_4}\right]}\right] \\ &= R_{\mathrm{ct}} + \frac{1}{2F}\left[\frac{\frac{b_3}{a_2}(j\omega) + \left(\frac{b_2}{a_2} - \frac{b_4}{a_4}\right)(j\omega)^{1/2} + \frac{b_1}{a_2}}{(j\omega)^{1/2}\left[(j\omega)\frac{a_4}{a_2} + 1\right]}\right]\end{aligned} \tag{6.27}$$

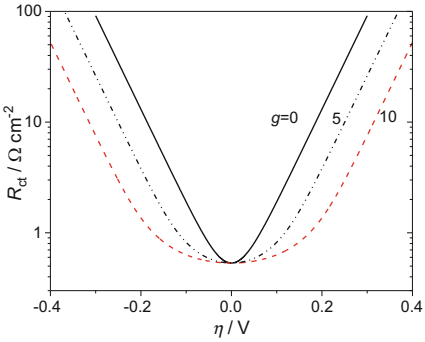
p. 152, Eq. (6.28)

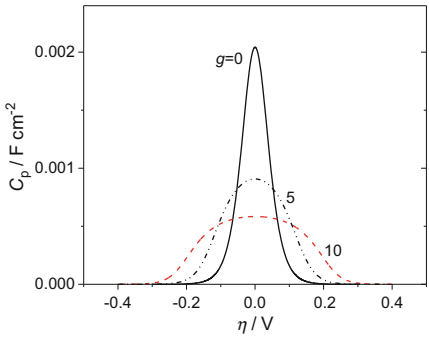
$$\begin{aligned}\widehat{Z}_{\mathrm{f}} &= R_{\mathrm{ct}} + \frac{1}{j\omega C_{\mathrm{p}} + \frac{1}{R_{\mathrm{p}}}} + Z_{\mathrm{W}} = R_{\mathrm{ct}} + \frac{1}{j\omega C_{\mathrm{p}} + \frac{1}{R_{\mathrm{p}}}} + \frac{\sigma}{\sqrt{j\omega}} \\ &= R_{\mathrm{ct}} + \frac{(j\omega)R_p C_p \omega + R_p (j\omega)^{1/2} + \sigma}{(j\omega)^{1/2}\left[(j\omega)R_p C_p + 1\right]}\end{aligned} \tag{6.28}$$

p. 153, Eq. (6.29)

$$\begin{aligned}&R_{\mathrm{ct}} = \frac{1}{2F}\frac{b_4}{a_4}; R_{\mathrm{p}} = \frac{1}{2F}\left(\frac{b_2}{a_2} - \frac{b_4}{a_4}\right); C_{\mathrm{p}} = 2F\frac{b_3}{b_1}\frac{1}{\left(\frac{b_2}{a_2} - \frac{b_4}{a_4}\right)}; \\ &\sigma = \frac{1}{2F}\frac{b_1}{a_2}; b_1 = \frac{a_2 b_3}{a_4}\end{aligned} \tag{6.29}$$

p. 157, Fig. 7.1, there is: "continuous line Langmuir isotherm, g = 1"
should be: "continuous line Langmuir isotherm, g = 0"
Corrected figure:





p. 160, Eqs. (7.19)–(7.20)

$$\begin{aligned} k_3 &= k_3^0 \Gamma_\infty^2 \\ k_{-3} &= k_{-3}^0 \Gamma_\infty^2 a_{H_2}^* \end{aligned} \tag{7.21}$$

p. 161, Eq. (7.23)

$$v_2 = v_2^0 \left[\begin{array}{l} \left(\dfrac{\theta_H}{\theta_H^*}\right)\left(\dfrac{a_{H_2O}}{a_{H_2O}^*}\right) \exp\left(-\beta_2 f \eta\right) \\ -\left(\dfrac{1-\theta_H}{1-\theta_H^*}\right)\left(\dfrac{a_{H_2}}{a_{H_2}^*}\right)\left(\dfrac{a_{OH^-}}{a_{OH^-}^*}\right) \exp\left[(1-\beta_2) f \eta\right] \end{array} \right] \tag{7.23}$$

p. 161, Eq. (7.27)

$$v_3^0 = \frac{k_3^0 k_{-3}^0 \Gamma_\infty^2 a_{H_2}^*}{\left(\sqrt{k_3^0} + \sqrt{k_{-3}^0 a_{H_2}^*}\right)^2} = \frac{k_3 k_{-3}}{\left(\sqrt{k_3} + \sqrt{k_{-3}}\right)^2} \tag{7.27}$$

p. 161, Eq, (7.30)

$$\frac{k_1 k_2}{k_{-1} k_{-2}} = \frac{k_1^2 k_3}{k_{-1}^2 k_{-3}} = \frac{k_{-2}^2 k_3}{k_2^2 k_{-3}} = 1 \tag{7.30}$$

p. 193, Eq. (8.28)

$$\widehat{C} = \frac{1}{j\omega\left(\widehat{Z}_{tot} - R_s\right)} C_{dl} + \frac{1}{\frac{1}{C_{ad}} + \sigma_{ad}\sqrt{j\omega} + R_{ad}} \tag{8.28}$$

p. 205, l. 8

$$\widehat{Z}_s = 1/(j\omega C_{dl} 2\pi r l)$$

p. 207, Fig. 9.3, legend.

$$C_{\mathrm{dl}} = 20\,\mu\mathrm{F\,cm}^{-2}$$

p. 338

There is:

$$\mathrm{Hg} \mid \mathrm{Hg_2SO_4} \mid \mathrm{SO_2}^{2-}$$

Should be:

$$\mathrm{Hg} \mid \mathrm{Hg_2SO_4} \mid \mathrm{SO_4}^{2-}$$